

BEST PRACTICES

WAFT | Guidelines for Successful Evolution

Project Setup

DO

- ✓ Use `waft new` to create projects
- ✓ Run `waft verify` before starting
- ✓ Configure version control (git)
- ✓ Set up checkpointing early

DON'T

- ✗ Manually create directory structure
- ✗ Skip verification steps
- ✗ Run evolution without backups
- ✗ Ignore Flight Recorder setup

Agent Design

DO

- ✓ Start with simple base agents
- ✓ Mark mutable regions clearly
- ✓ Add error handling
- ✓ Test agents before evolution

DON'T

- ✗ Start with complex agents
- ✗ Hard-code everything
- ✗ Skip validation
- ✗ Ignore Scint types

Evolution Configuration

DO

- ✓ Start with small populations (10-20)
- ✓ Use conservative mutation rates (0.1)
- ✓ Enable elitism (preserve best 2)
- ✓ Run many generations (50+)

DON'T

- ✗ Start with huge populations
- ✗ Use extreme mutation (>0.3)
- ✗ Disable elitism entirely
- ✗ Expect results in 5 generations

Monitoring

DO

- ✓ Check fitness trends regularly
- ✓ Review Flight Recorder logs
- ✓ Watch for plateaus
- ✓ Analyze mutation impact

DON'T

- ✗ Run evolution blind
- ✗ Ignore declining fitness
- ✗ Let plateaus continue indefinitely
- ✗ Skip post-evolution analysis

Scientific Rigor

WAFt produces scientific data. Treat it accordingly:

1. **Document everything** — Use Empirica sessions
2. **Control variables** — Change one thing at a time
3. **Replicate results** — Run experiments multiple times
4. **Share findings** — Contribute to the community